



Tire Development Instruction Package NV56 – Base Summer – 19 Square

255/45R19 104W XL

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TECHNICAL SPECIFICATION #:
REVISION #:

TS-1188251
03

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DOCUMENT HISTORY

Rev	Date	Description	Author
01	18APR2018	Initial Release	Guzman
02	20APR2018	Added: Changed: 2.5 Tire Profile Reporting – Loaded Parameters. Removed:	Guzman
03	14MAR2019	Minor updates: IPLR, Flat spotting	HAVER

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1. Scope and Objective

This document provides attributes for performance and durability validation of Tesla part number:

1188251-XX-X

255/45R19 104 W XL

Any information not contained in this specification, but listed separately on other Tesla Motors official documentation (such as part drawing, etc.) shall also be satisfied.

2. General Specification

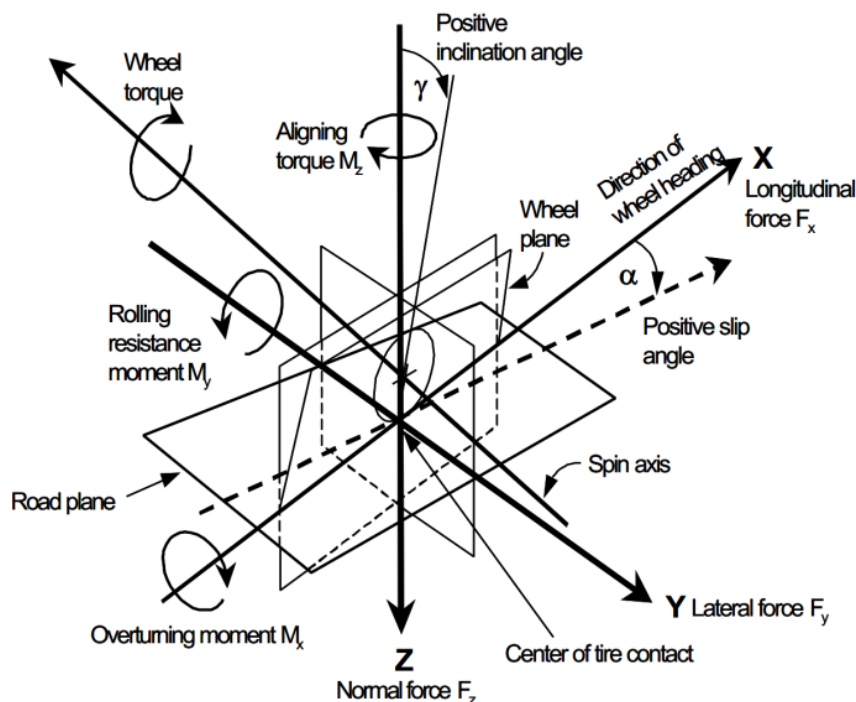
2.1. Terms and Definitions

Throughout this specification, various terms are used and their definitions must be understood for proper interpretation of this document. These terms and definitions are listed below.

Term	Definition
F_Y	Force in the Y direction, also called Lateral Force
F_X	Force in the X direction, also called Longitudinal Force
M_Z	Moment about the Z axis
SA	Slip Angle is the angle created between the path that the tire is traveling and the direction the tire is pointing.
SR	Slip Ratio is the ratio of the longitudinal travel of the wheel divided by the travel of the tire.
SLR	Static Load Radius is the radius of the tire measured from the center of the tire to the ground plane at a specific vertical load while the vehicle is no moving.
ERR	Equivalent Rolling Radius is the radius of the tire measured from the center of the tire to the ground plane while rotating at a certain speed and at a specific vertical load. Typically ERR is back calculated from revolutions per distance traveled.
PRAT	Plysteer Residual Aligning Torque
PRCF	Plysteer Residual Cornering Force

2.2. Coordinate System Definition

All forces and moments follow the SAE tire axis system as defined in SAE J2047.



2.3. Vehicle Information

Summary Vehicle Information relevant to tire application is listed below.

Item Name	Units	Data Point Front	Data Point Rear	Note
Applied Tire Pressure	kPa	290	290	Tuning Item
Tire Load - Curb	kg	525	550	-
Tire Load – Performance	kg	560	580	2 Up
Tire Load - GVW	kg	600	725	-
Tire Load – GVW + Towing	kg	600	815	-
Top Speed	kph	225		-
Tire Life	mi	50K	50K	Rotation Allowed
Peak Power	W	-		-
Application Rim Size	inch	19x9.5J	19x9.5J	
Acoustic Foam	-	Yes	Yes	Density Tuning Item
Sensor on Tire	-			None
Application Camber	Deg	-0.5	-1.0	Tuning Item
Application Toe	Deg	0	0.2IN	Tuning Item
Tire Size	-	255/45R19	255/45R19	
Load Index	-	104	104	
Speed Rating	-	W	W	
Tesla Marking	-	T0	T0	Etched or Slugged
Maximum Factory New Tire Age	Weeks	26	26	
Maximum Service New Tire Age	Weeks	52	52	

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2.4. Tire Construction Reporting

Each unique tire construction shall have a corresponding Tire Construction Sheet. Design Intent, Tread Details, and Construction Details from template below to be provided before each Submission as defined in Section 3. Additional data to be provided for Preferred Constructions as defined in Section 3.

Table 1: Tire Construction Sheet. Template can be provided by Tesla Chassis Engineering

		Submission # 1		
		1	2	3
Constr. Code		123456A	123456B	123456C
Tire supplier's description of design intent for each construction		Regular	Lowest RRC	
Tread				
Tread Compound		123456	←	←
Under-Tread Compound		789123	←	←
Construction Details				
Overlay material		140/1	←	←
Overlay design (center, shoulder, angle)		Classic	←	←
Belt material & type		Steel - Hard Rubber 4x30	Steel - Soft Rubber 4x30	←
Top belt cured angle		+/- 26°	←	←
Sidewall Material		2 ply	←	←
Body ply material		Polyester	←	←
Bead matrix/design		2 piece	←	←
Apex/bead filler material		Hard/Hard	←	Soft/Soft
Apex/bead filler height (mm)		60	←	←
Tire Dimensions & Mechanics				
Tire mass (kgs)				
Outer Diameter (mm)				
Section Width (mm)				
Tread depth - center (mm)			←	←
Tread depth - shoulder (mm)			←	←
Footprint (area, shape, width ratio)		Picture w/ void ratio within footprint		
RPK		report in 2D table as specified in TDIP		
DLR		report in 2D table as specified in TDIP		
SLR (70% of Load) (mm)	Front Axle			
SLR (100% of Load) (mm)				
ERR (100% Load) (mm)				
SLR (70% of Load) (mm)	Rear Axle			
SLR (100% of Load) (mm)				
ERR (100% Load) (mm)				
Rolling Resistance				
RRc - SAEJ2452 Coefficients	α			
	β			
	a			
	b			
	c			
RRc - SAEJ2452 (kg/ton) (@ TDIP Conditions)				
Force and Moment (MF 6.2 with SWIFT)				
Cornering Stiffness (Normalized Fy/SA)		.tir file Report Criteria from TDIP		
Tractive Force (Normalized Fx/SA)				
Aligning Torque Stiffness (Normalized Mz/SA)				
XYZ Stiffness (N/mm)				
Dynamic Stiffness (Relaxation Length, m/SA)				
PRAT (Nm) *Measured at PPAP				
Noise Vibration and Harshness				
Max Sound Pressure Level (dba at frequency)				
Max Structural Vibration (g at frequency)				

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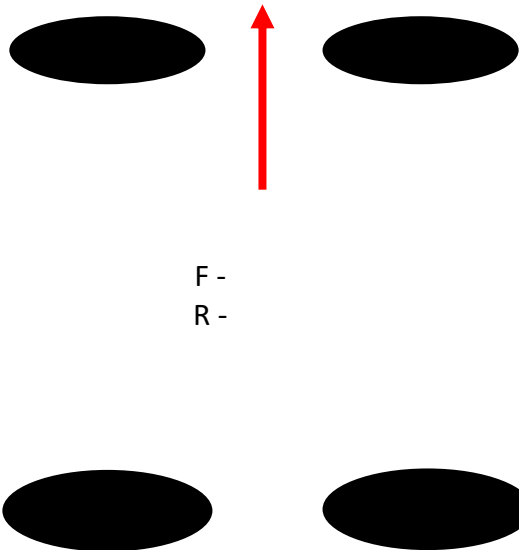
2.5. Tire Profile Reporting

Each preferred constrictor as defined in Section 3, shall have a corresponding Tire Profile using analysis tools with design intent constructions and parameters.

	Vertical Load	Centrifugated Speed	Toe	Camber
	kg	kph	Deg	deg
Loaded Front	525	140	0.05 OUT	-0.5
Loaded Rear	550	140	0.20 IN	-1.0
Inflated F	0	0	0	0
Inflated R	0	0	0	0

2.6. Tread Pressure Distribution Reporting

Each preferred constrictor as defined in Section 3, shall have a corresponding Tread Pressure Distribution Reporting using analysis tools with design intent constructions and parameters. & Confirm Before Hardware Order.



		Vertical Load	Lateral Load	Longitudinal Load	Pressure	Speed	Slip Angle	Slip Ratio	Camber Angle
		kg	N	N	kPa	kph	Deg	%	Deg
Condition 1	Front	5150	0	0	303	0	0	0	-0.5
	Rear	5400	0	0	303	0	0	0	-1.0
Condition 2	Front	7700	6160	0	303	80	3.2	0	+2.08
	Rear	8100	6480	0	303	80	2.6	0	+1.51
Condition 3	Front	4000	0	2000	303	20	0	3	0
	Rear	6500	0	2000	303	20	0	3	-2.0

Conditions represent Hot Pressures

2.7. Virtual Tire Reporting

Each proposed construction shall have a virtual tire model submitted to Tesla Engineering before constructions are built. Virtual submission shall satisfy the format for Equation 2.6.1 for a given Normal Load. Normal load sensitivities for each Virtual Coefficient shall be provided through second order polynomial curve fits for each coefficient. Target values for each coefficient shall be set by Tesla Engineering.

$$F_y = D * \sin[C \arctan\{B\alpha - E (B\alpha - \arctan(B\alpha))\}] \quad (2.6.1)$$

Table 2: Pacejka MF simplified coefficients

Name	Coefficient	Unit	Equivalent	Requirement	Target
Lateral Force	F_y	N	F_y	Output	
Slip Angle	α	deg	α	Variable	
Cornering Stiffness	BCD	$\frac{N}{deg}$	C_α	Virtual	4.2.1
Peak Cornering	D	-	μ	Virtual	4.2.1
Shape Factor	C	-	*	Virtual	-
Stiffness Factor	B	-	$\frac{C_\alpha}{C * D}$	Virtual	4.2.1
Curvature Factor	E	-	**	Virtual	-
Vertical Stiffness	K_z	$\frac{N}{mm}$	-	Virtual	4.2.4
Lateral Stiffness	K_y	$\frac{N}{mm}$	-	Virtual	4.2.5

* - Shape Factor [C] – usually constant function of F_z .
 ** - Curvature Factor [E] – usually a second order polynomial function of F_z .

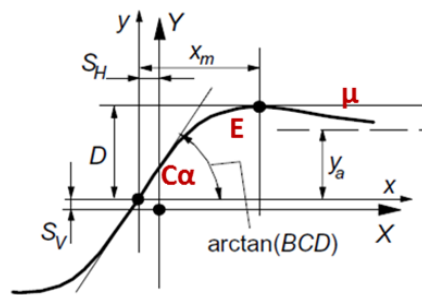


Figure 1: Proposed Virtual Output, graphical representation.

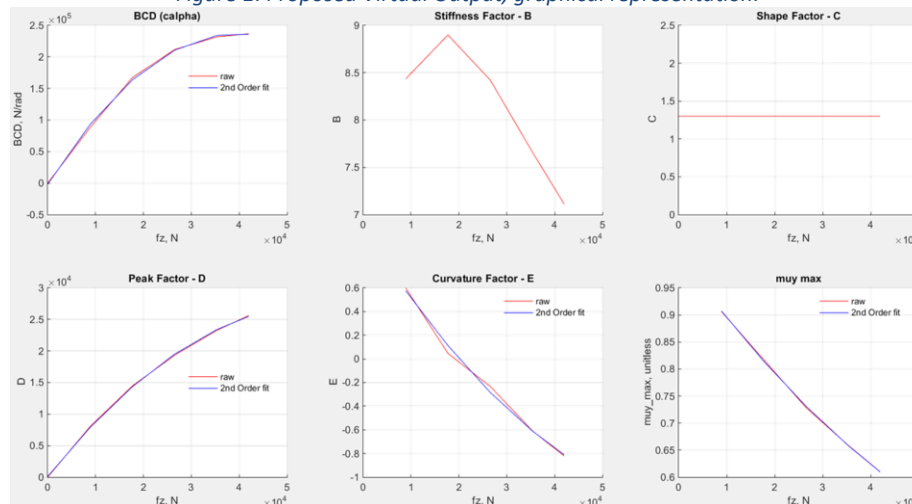


Figure 2: Normal Load Sensitivities for each Coefficient

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3. Confirmation Item Summary

S# (Submission No.) tests are intended to develop the subjective rating correlation to various design engineering decisions. There may be multiple submissions to allow for iterations in component design.

PV (Production Validation) tests shall be completed using initial parts from production tooling and process and are intended to identify relationships between design and process characteristics.

IP (In-Process) tests shall be completed with production parts on an ongoing basis. Supplier shall be responsible for creating and maintaining Control Plan documentation to establish a basis for continuous improvement.

Results of all reports shall be presented in DVP&R and other more detailed format including pictures of test setups, actual test parts, and detailed tables of observations.

The following table summarizes the minimum requirements for DV and PV testing. Supplier shall propose additional tests where applicable to further guarantee life and performance for the entirety of the vehicle life.

Item #	Item Name	Test Requirement		
<u>4.1 Dimensional</u>		S#	PV	IP
4.1.1	Width	A	•	Control Plan
4.1.2	Overall Diameter	A	•	Control Plan
4.1.3	Mass	A	•	Control Plan
4.1.4	Moment of Inertia	B	•	Control Plan
4.1.5	Static Loaded Radius	A	•	Control Plan
4.1.6	Dynamic Loaded Radius	C	•	Control Plan
4.1.7	Revolutions per kilometer	C	•	Control Plan
<u>4.2 Force and Moment (SWIFT MF6.2)</u>				
4.2.1	Lateral Force Performance	A	•	Control Plan
4.2.2	Aligning Torque	A	•	Control Plan
4.2.3	Tractive Force	B	•	Control Plan
4.2.4	Vertical Stiffness (N/mm)	A	•	Control Plan
4.2.5	Lateral Stiffness (N/mm)	A	•	Control Plan
4.2.6	Plysteer Residual Aligning Torque	C	•	Control Plan
4.2.7	Plysteer Residual Cornering Force	C	•	Control Plan
4.2.8	Vertical Damping	C	•	Control Plan
4.2.9	Dynamic Stiffness, Relaxation Length (m/SA)	A	•	Control Plan
4.2.10	Snow Traction	C	•	Control Plan
<u>4.3 Rolling Resistance</u>				
4.3.1	RR _c (SAE J2452 method)	A	•	Control Plan
<u>4.4 Noise Vibration & Harshness</u>				
4.4.1	Max Sound Pressure Level	B	•	Control Plan
4.4.2	Max Structural Vibration (g at frequency)	B	•	Control Plan
4.4.3	Vehicle Transmissibility	A	•	Control Plan
<u>4.5 Tread-wear</u>				
4.5.1	Fastest wearing groove (mm/10,000 miles)	C	•	Control Plan
4.5.2	Average wear rate (mm/10,000 miles)	C	•	Control Plan
4.5.3	Irregular wear rating (mm/10,000 miles)	C	•	Control Plan
<u>4.6 Subjective Handling</u>				
4.6.1	Complete Table	•	•	Control Plan
<u>4.7 Uniformity</u>				
4.7.1	Conicity	C	•	Control Plan
4.7.2	Radial RunOut	C	•	Control Plan
4.7.3	Radial Force Peak to Peak	C	•	Control Plan
4.7.4	Radial First Harmonic	C	•	Control Plan
4.7.5	Lateral Force Peak to Peak	C	•	Control Plan
4.7.6	Lateral RunOut	C	•	Control Plan
4.7.7	Inflation Pressure Loss Rate	A	•	Control Plan
4.7.8	Flat Spotting – Short Term	B	•	Control Plan
4.7.9	Flat Spotting – Long Term	B	•	Control Plan

- A – Test to be completed for each construction, before submission. Virtual or simulation data OK.
 B – Test to be completed for preferred construction, after submission
 C – Test to be completed for preferred construction, after final submission
 • – Test to be completed for each construction, during submission.

4. Confirmation Items

Tested components are to be of random sampling and not repeat the same tests, unless otherwise instructed by Tesla Engineering.

4.1. Dimensional

Item #	Item Name	Pressure (kPa)	Unit	Data Front	Data Rear
4.1.1	Overall Width	290	mm	252 ± 3	252 ± 3
4.1.2	Overall Diameter	290	mm	719 ± 3	719 ± 3
4.1.3	Mass	-	kg	10.8 Max	10.8 Max
4.1.4	Moment of Inertia	-	kg*m ²	I _{xx}	TBC
		-		I _{yy}	TBC

4.1.5. Static Loaded Radius

SLR (mm)		Pressure (kPa)		
Mass (kg)		276	290	310
Curb Front	525	340 ± 5	340 ± 5	340 ± 5
Curb Rear	550	340 ± 5	340 ± 5	340 ± 5
Greatest Vehicle Axle	725	335 ± 5	335 ± 5	335 ± 5

4.1.6. Dynamic Loaded Radius

Report the dynamic loaded radius for the following speeds at the application tire pressure:

DLR (mm)		Speed (kph)				
Mass (kg)		15	80	130	195	240
Performance Front	560					
Performance Rear	580					
Greatest Vehicle Axle	725					

4.1.7. Revolutions per kilometer

Report revolutions per kilometer for the following speeds at the application tire pressure:

rev/km		Speed (kph)				
Mass (kg)		15	80	130	195	240
Performance Front	560					
Performance Rear	580					
Greatest Vehicle Axle	725					

$$ERR = \frac{\text{distance traveled}}{\text{revolutions} * 2\pi}$$

$$ERR = \frac{1 \text{ km}}{2\pi * 500 \text{ revs}} = .318 \text{ m} = 318 \text{ mm}$$

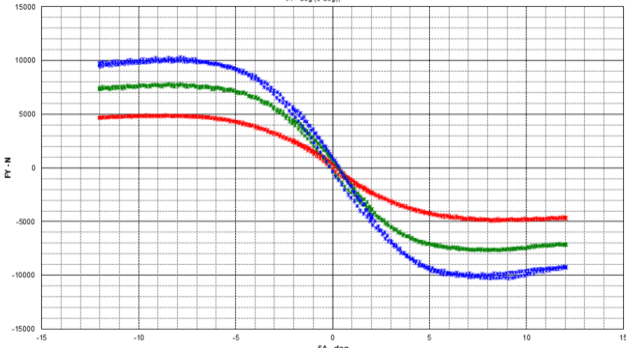
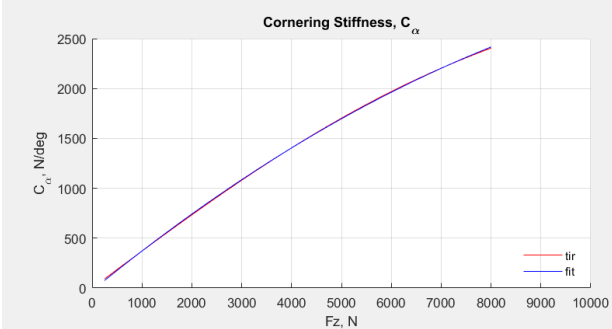
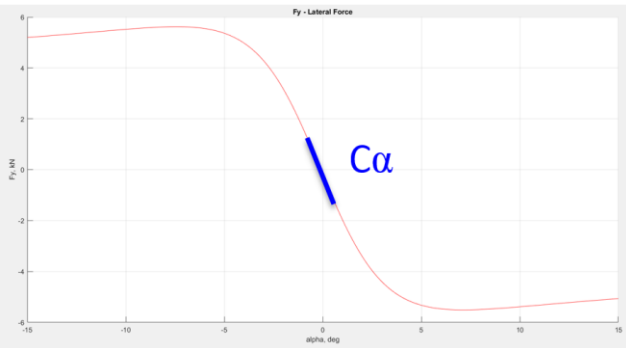
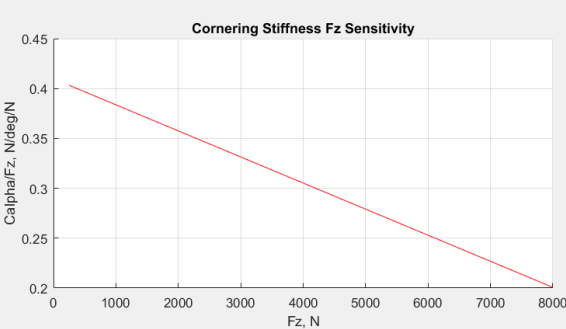
4.2. Force Moment

As measured by a force and moment, the output of the data should be in **MF6.2 format**, and adequate to create the below characterization plots. Data analysis will be performed by Tesla Motors Inc.

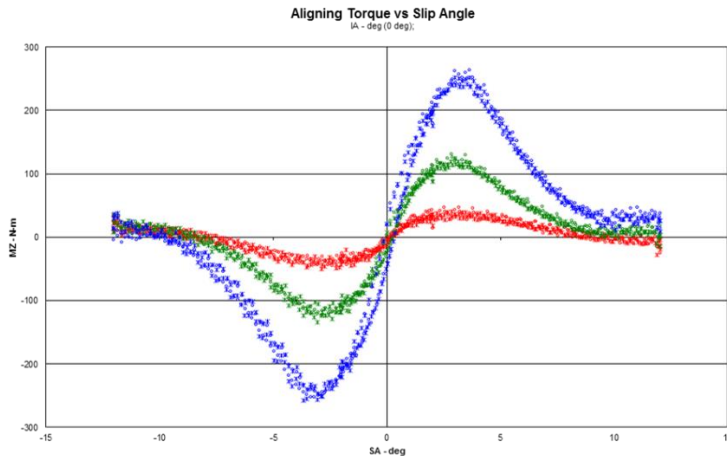
Load values, pressure, and resolution should be in accordance with standards and practices at Smithers Rapra to Tesla Protocol. Collected F&M data will be available in the .tir file for each dimension.

Criteria values shall be reported normalized to Reference Tire in 4.6. Measured reference tire shall follow similar test method as submission data at Smithers Rapra.

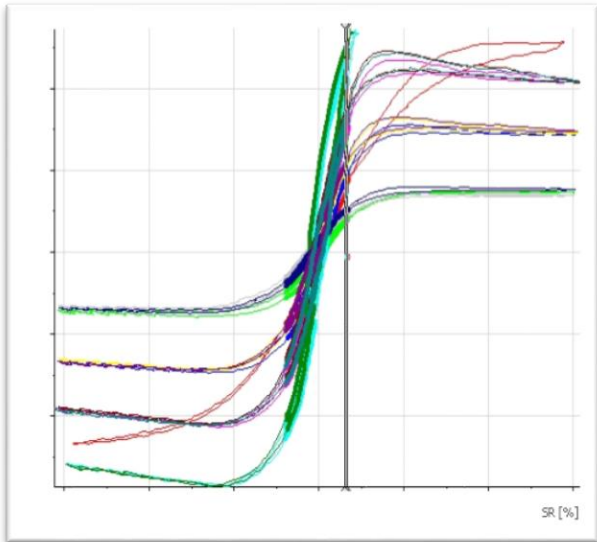
4.2.1. Lateral Force Performance

Description	Cornering Coefficient is the force in the Y direction divided by the Slip Angle and normalized by vertical load.																								
Set Up	The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.																								
Image	Lateral Force vs Slip Angle (α - deg (0 deg))  Cornering Stiffness, C_{α}  Fy - Lateral Force  Cornering Stiffness Fz Sensitivity 																								
	Criteria	<table><thead><tr><th>Attribute</th><th>Parameters</th><th>Target</th><th>Units</th><th>Normalized to Reference</th></tr></thead><tbody><tr><td>Cornering Stiffness Coefficient</td><td>SA = 0 Fz = 0</td><td>.35 MIN</td><td>(N/deg)/N</td><td>0.85 MIN</td></tr><tr><td>Cornering Stiffness</td><td>Fz = 5.5 kN</td><td>1850 ± 5%</td><td>N/deg</td><td>1.05</td></tr><tr><td>Normalized Lateral Force</td><td>Fz = 5.7 kN</td><td>TBD</td><td>-</td><td>TBD</td></tr></tbody></table> Report Comparison to Reference Tire in 4.6					Attribute	Parameters	Target	Units	Normalized to Reference	Cornering Stiffness Coefficient	SA = 0 Fz = 0	.35 MIN	(N/deg)/N	0.85 MIN	Cornering Stiffness	Fz = 5.5 kN	1850 ± 5%	N/deg	1.05	Normalized Lateral Force	Fz = 5.7 kN	TBD	-
Attribute	Parameters	Target	Units	Normalized to Reference																					
Cornering Stiffness Coefficient	SA = 0 Fz = 0	.35 MIN	(N/deg)/N	0.85 MIN																					
Cornering Stiffness	Fz = 5.5 kN	1850 ± 5%	N/deg	1.05																					
Normalized Lateral Force	Fz = 5.7 kN	TBD	-	TBD																					

4.2.2. Normalized Aligning Torque

Description	Aligning torque stiffness is the moment about the z axis divided by the Slip Angle and normalized by vertical load.																								
Set Up	The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.																								
Image																									
Criteria	<table><tr><th>Attribute</th><th>Parameters</th><th>Target</th><th>Units</th><th>Normalized to Reference</th></tr><tr><td>Aligning Torque Stiffness</td><td>Fz = 5.5 kN</td><td>90 ± 10%</td><td>Nm/deg</td><td>1.14</td></tr><tr><td>Pneumatic Trail</td><td>Fz = 5.7 kN SA @ F_{y, MAX}</td><td>TBD</td><td>m</td><td>-</td></tr><tr><td>Peak Aligning Moment</td><td>Fz = 5.7 kN</td><td>TBD</td><td>Nm</td><td>-</td></tr></table>					Attribute	Parameters	Target	Units	Normalized to Reference	Aligning Torque Stiffness	Fz = 5.5 kN	90 ± 10%	Nm/deg	1.14	Pneumatic Trail	Fz = 5.7 kN SA @ F _{y, MAX}	TBD	m	-	Peak Aligning Moment	Fz = 5.7 kN	TBD	Nm	-
Attribute	Parameters	Target	Units	Normalized to Reference																					
Aligning Torque Stiffness	Fz = 5.5 kN	90 ± 10%	Nm/deg	1.14																					
Pneumatic Trail	Fz = 5.7 kN SA @ F _{y, MAX}	TBD	m	-																					
Peak Aligning Moment	Fz = 5.7 kN	TBD	Nm	-																					
Report comparison to Reference Tire in 4.6																									

4.2.3. Normalized Tractive Force

Description	Tractive force is the force in the X direction divided by the slip ratio and normalized by vertical load.				
Set Up	The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.				
Image					
Criteria	To be confirmed Report comparison to Reference Tire in 4.6				

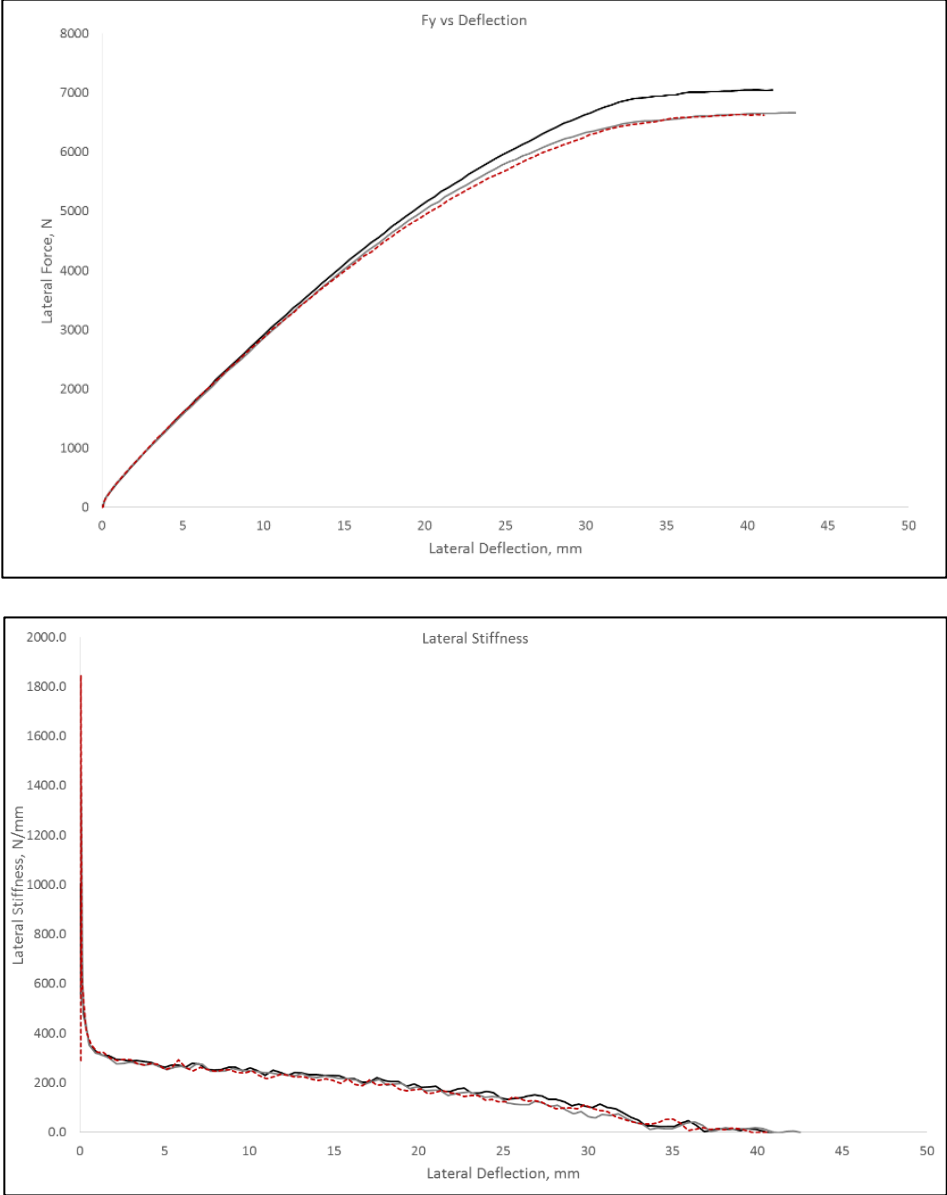
4.2.4. Vertical Stiffness

Description	Vertical stiffness is the slope of the vertical load verses tire radius plot.
Set Up	The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.
Parameters	Pressure [kPa]: 276,290,310
Image	
Criteria	305 N/mm $\pm$ 10% @ Application Pressure Report comparison to Reference Tire in 4.6

4.2.5. Vertical Damping

Description	Measure the vertical damping ratio of a free bouncing tire.
Set Up	Supplier Test Method. TBD.
Image	
Criteria	To be confirmed

4.2.6. Lateral Stiffness

Description	Lateral stiffness is the slope of the lateral load verses tire deformation plot.
Set Up	The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.
Parameters	Pressure [kPa]: 276,290,310
Image	 The image contains two plots. The top plot, titled 'Fy vs Deflection', shows Lateral Force, N on the y-axis (0 to 8000) versus Lateral Deflection, mm on the x-axis (0 to 50). It features three curves (black, grey, and red) that start at the origin and increase with a decreasing slope, reaching approximately 7000 N at 40 mm deflection. The bottom plot, titled 'Lateral Stiffness', shows Lateral Stiffness, N/mm on the y-axis (0.0 to 2000.0) versus Lateral Deflection, mm on the x-axis (0 to 50). It features three curves (black, grey, and red) that start at a high stiffness value (around 1800 N/mm) at 0 mm deflection and rapidly decrease, leveling off around 200 N/mm at 40 mm deflection.
Criteria	Measured at Application Pressure: Lateral Stiffness ($K_{Y,1}$) at 5 mm deflection: 320 N/mm $\pm$ 5% Lateral Stiffness at 20mm = 0.55 * $K_{Y,1}$ MIN First derivative reported. Report comparison to Reference Tire in 4.6

4.2.7. Plysteer Residual Aligning Torque

Description	Plysteer Residual Aligning Torque is the moment about the Z-axis when lateral force is zero.
Set Up	The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.
Parameters	Load [kg]: 525,580 Pressure: Application pressure
Image	The graph shows two linear relationships. The red line represents Cornering Force (Fy) in Newtons, starting at 300 N at 0 degrees slip angle and decreasing to -300 N at 0.2 degrees. The orange line represents Aligning Moment (Mz) in Newton-meters, starting at -300 Nm at 0 degrees and increasing to 300 Nm at 0.2 degrees. The intersection point, labeled PRAT, occurs at a slip angle of approximately 0.125 degrees and an aligning moment of -5 Nm.
Criteria	$-2.0 \pm 0.5 \text{ Nm}$

4.2.8. Plysteer Residual Cornering Force

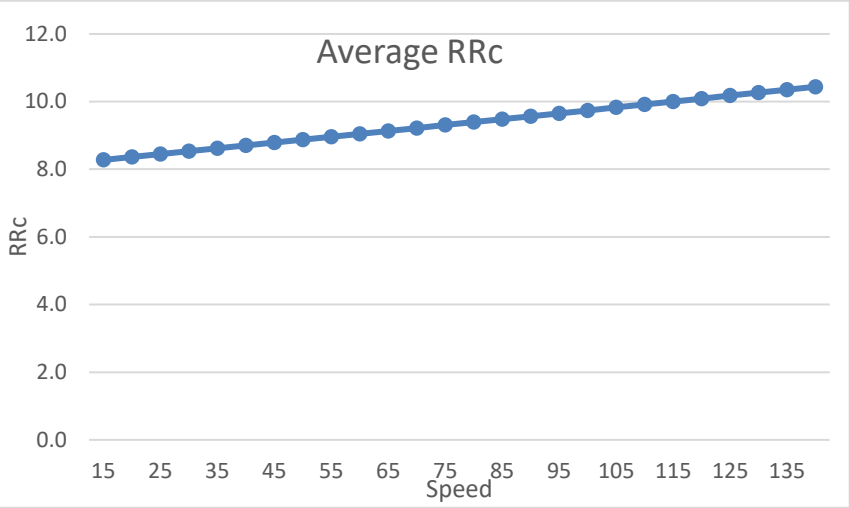
Description	Plysteer Residual Cornering Force is a force in the Y-Axis when Aligning Moment is 0.
Set Up	The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.
Parameters	Load [kg]: 525,580 Pressure: Application pressure
Image	The graph shows two linear relationships. The red line represents Cornering Force (Fy) in Newtons, starting at 300 N at 0 degrees slip angle and decreasing to -300 N at 0.2 degrees. The orange line represents Aligning Moment (Mz) in Newton-meters, starting at -300 Nm at 0 degrees and increasing to 300 Nm at 0.2 degrees. The intersection point, labeled PRCF, occurs at a slip angle of approximately 0.125 degrees and an aligning moment of -5 Nm.
Criteria	Report when aligning moment is zero.

4.2.9. Relaxation Length (m/SA)

Description	Measure the relaxation length of the tire.
Set Up	Mathematically acquired from Cornering Stiffness and Lateral Stiffness $L_R = \frac{C_\alpha}{K_Y} * \left(\frac{360^\circ}{2\pi} \right)$ $mm = \frac{N/^\circ}{N/mm} * \left(\frac{360^\circ}{2\pi \text{ RAD}} \right)$
Image	
Criteria	R _L = 360 MAX at Fz = 5.7 kN

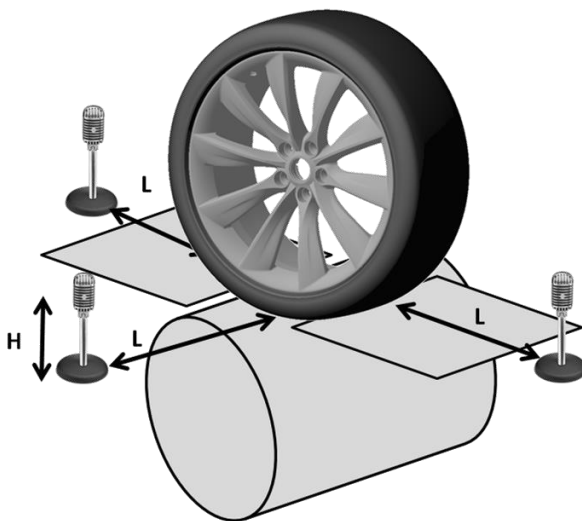
4.3. Rolling Resistance

4.3.1. Rolling Resistance – SAE

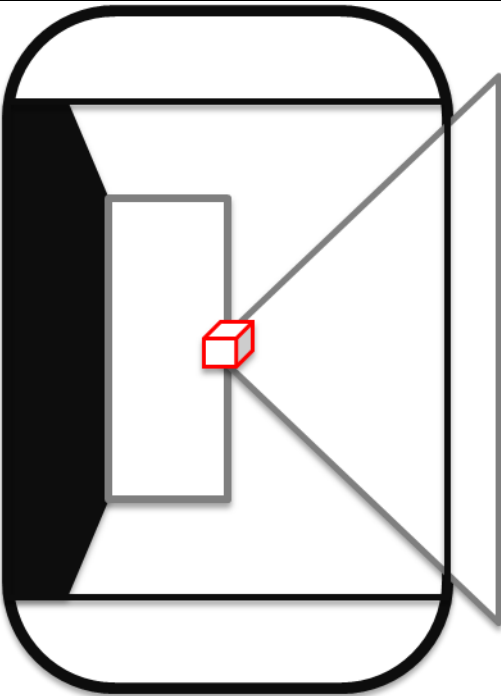
Description	Measure the effort required to keep a tire rolling.																												
Set Up	The rolling resistance coefficient should follow SAE J2452, with Tesla Parameter Modifications. $RR=F/P=L^b(a+bv+cv^2)$ RR Coefficient = 1000 (RR/L) Units: RR (N), L (N)																												
Parameters	Pressure Sweep [kPa]: 250,280,310 Speed Sweep [kph]: 115 - 15 Load [% of Max Load Capacity]: 70,85,100 Samples: 2 per S# construction. 4 additional per preferred construction For Split Fits, report RRC individually and as Vehicle Average.																												
Image	 <table border="1"> <caption>Approximate data points from the 'Average RRC' graph</caption> <thead> <tr> <th>Speed (kph)</th> <th>RRC</th> </tr> </thead> <tbody> <tr><td>15</td><td>8.2</td></tr> <tr><td>25</td><td>8.4</td></tr> <tr><td>35</td><td>8.6</td></tr> <tr><td>45</td><td>8.8</td></tr> <tr><td>55</td><td>9.0</td></tr> <tr><td>65</td><td>9.2</td></tr> <tr><td>75</td><td>9.4</td></tr> <tr><td>85</td><td>9.6</td></tr> <tr><td>95</td><td>9.8</td></tr> <tr><td>105</td><td>10.0</td></tr> <tr><td>115</td><td>10.2</td></tr> <tr><td>125</td><td>10.4</td></tr> <tr><td>135</td><td>10.5</td></tr> </tbody> </table>	Speed (kph)	RRC	15	8.2	25	8.4	35	8.6	45	8.8	55	9.0	65	9.2	75	9.4	85	9.6	95	9.8	105	10.0	115	10.2	125	10.4	135	10.5
Speed (kph)	RRC																												
15	8.2																												
25	8.4																												
35	8.6																												
45	8.8																												
55	9.0																												
65	9.2																												
75	9.4																												
85	9.6																												
95	9.8																												
105	10.0																												
115	10.2																												
125	10.4																												
135	10.5																												
Criteria	RRc: 7.1 ± 0.20 Vehicle Average [Application Pressure, 80 kph, 100% GVW load] Report: α, β, a, b, c coefficients for each tire tested Report comparison to Reference Tire in 4.6																												

4.4. Noise, Vibration, and Harshness

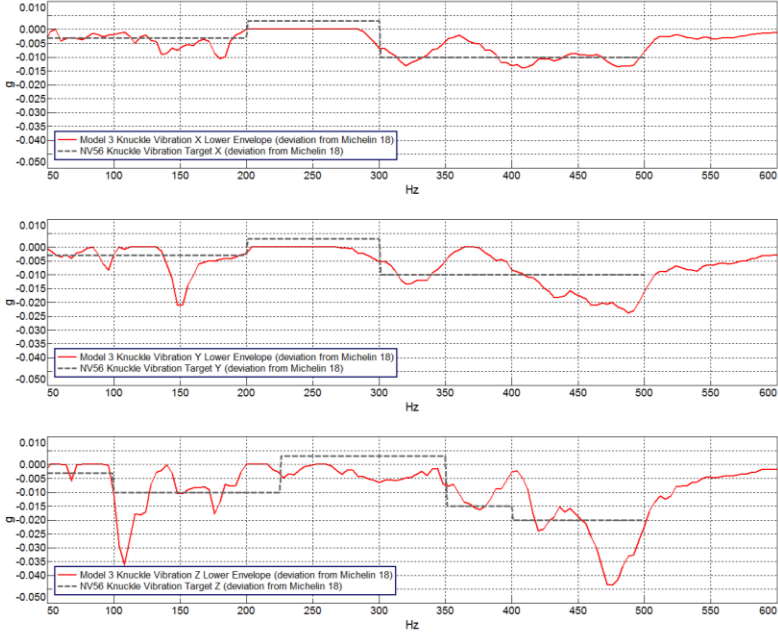
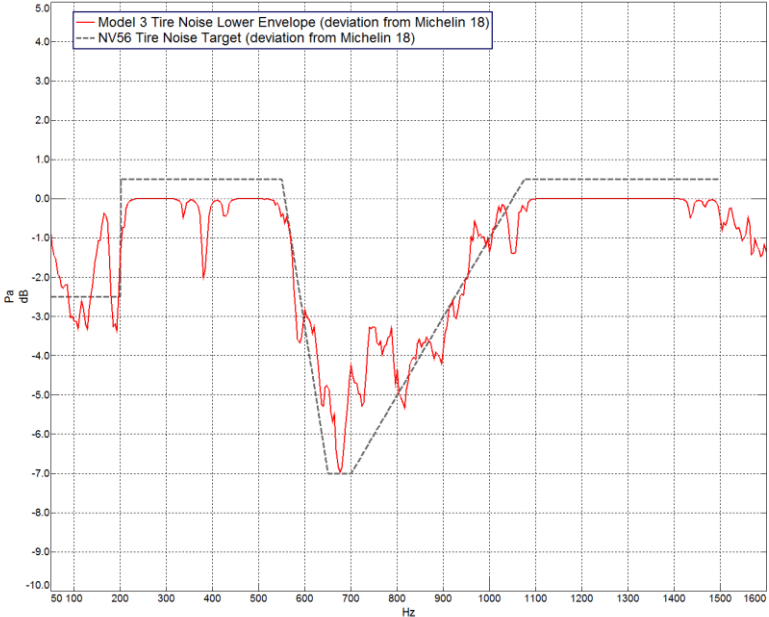
4.4.1. Sound Pressure Level

Description	Measure the tire road noise.
Set Up	- In an acoustic chamber - measure the sound pressure level at the entrance, exit, and side a distance away from the tire as shown in the picture below for a duration of 10s. Use both smooth and rough surfaces and at 50, 80, and 100 kph constant vehicle speeds. - Also, perform a coast down from 120 to 25 kph on both smooth and rough surfaces for duration 60s.
Parameters	Load [kg]: 580 Pressure [kPa]: 276,290 L [m]: 1 H [m]: 0.2
Image	 The diagram shows a tire mounted on a test rig. Three microphones are positioned around the tire: one at the front, one at the side, and one at the rear. The distance from the center of the tire to each microphone is labeled 'L'. The height of the microphones from the ground is labeled 'H'. The tire is shown on a cylindrical base, and the ground is represented by a horizontal plane.
Criteria	Data output FFT: 20 Hz to 20 kHz with a delta f of 4 Hz, Hanning Window, 50% overlap - Color map (time-frequency), Linear weighting (non A-weighting) - OAL verse time (non A-weighting) Report comparison to Reference Tire in 4.6

4.4.2. Structural Vibration

Description	Measure vibration of the bearing hub with a force transducer in each axis.
Set Up	- In an acoustic chamber - measure the vibration of the bearing hub in each axis (X,Y,Z) for a duration of 10s. Use both smooth and rough surfaces and at 50, 80, and 100 kph constant vehicle speeds. - Also, perform a coast down from 120 to 25 kph on both smooth and rough surfaces for duration 60s.
Parameters	Load [kg]: 580 Pressure [kPa]: 276,290
Image	
Criteria	Force Transducer: FX, FY, FZ Data output FFT: 0 Hz to 400 Hz with a delta f of 0.5 Hz, Hanning Window, 50% overlap - Color map (time-frequency) - OAL verse time Report comparison to Reference Tire in 4.6

4.4.3. Vehicle Transmissibility

Description	Measure noise and vibration of the tire when installed in vehicle.
Set Up	a. Measure vibrations on Front and Rear Knuckles at the Wheel Centers. b. Instrument vehicle with near-field microphone on trailing edge of contact patch for a Front and Rear tire.
Parameters	Load [kg]: Performance Mass per 2.3 Pressure [kPa]: Application Pressure per 2.3 Reference Tire: 235/45R18 Michelin MXM4 T0
Image	
Criteria	a. Relative to Reference Tire in 4.4.7, the following deviations are allowed between 50-500Hz.  b. Relative to Reference Tire in 4.4.7, the following deviations are allowed between 100-1500Hz when measured on the same pavement type. 

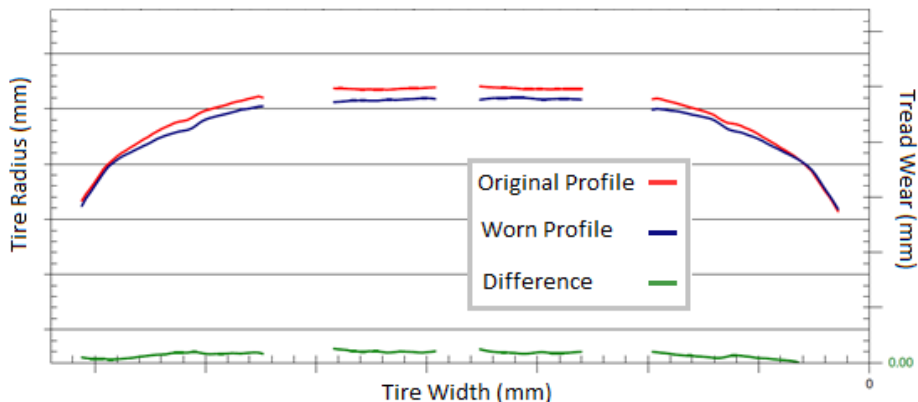
Printed copies are uncontrolled. Controlled documents reside electronically.

Individuals in possession of an uncontrolled copy are responsible for ensuring they are using the latest revision of this document.

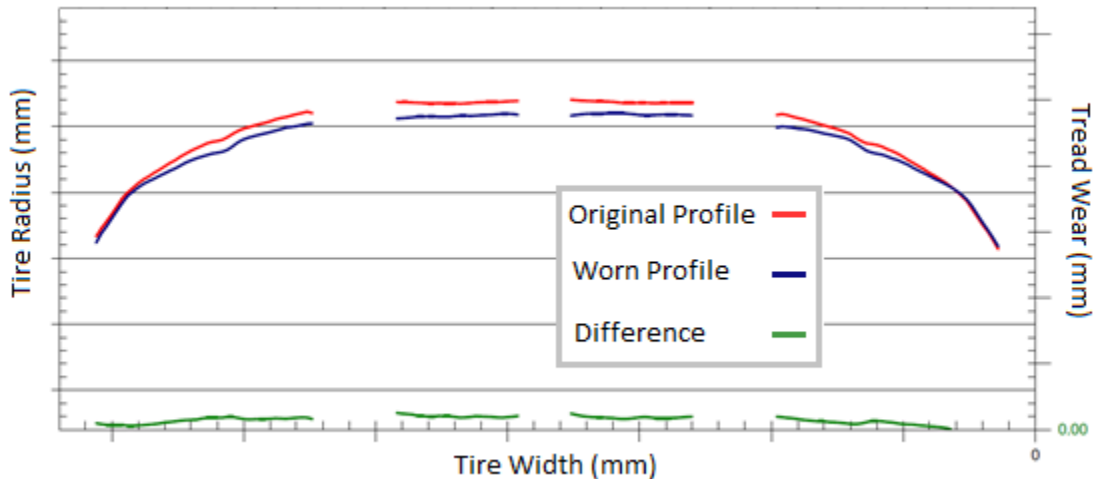
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4.5. Tire Wear

4.5.1. Fastest Wearing Groove

Description	Report the wear rate of the fastest wearing rib in mm per 10,000 miles.
Set Up	Supplier Test Method
Parameters	Vehicle Camber
Image	
Criteria	Worn Tire shall satisfy NVH Performance Criteria: 4.4.1 and 4.4.2 Fastest wearing groove shall be $\leq 15\%$ increase on Average Wear Rate 4.5.2

4.5.2. Average Wear Rate

Description	Report the average wear rate across the surface in mm per 10,000 miles
Set Up	Supplier Test Method
Parameters	Vehicle Camber.
Image	
Criteria	Worn Tire shall satisfy NVH Performance Criteria: 4.4.1 and 4.4.2

4.5.3. Irregular Wear Rate

Description	Report the wear rate across a rib that is different left to right of the tire center or across a single groove.
Set Up	Supplier Test Method
Parameters	Vehicle Camber
Image	
Criteria	Worn Tire shall satisfy NVH Performance Criteria: 4.4.1 and 4.4.2

4.6. Subjective Ride and Handling Rating

The subjective ride and handling rating should follow the industry standard 10 point rating system.

Vehicle and Attribute Customer Rating System

90% customer satisfaction imperative

Engineering Rating index	1	2	3	4	5	6	7	8	9	10
Evaluation of attribute performance	Not Acceptable		Poor		Borderline	Acceptable	Fair	Good	Very Good	Excellent
Customer satisfaction(for the attribute)	Very dissatisfied				Somewhat Dissatisfied	Fairly well satisfied		Very Satisfied	Completely satisfied	
Improvement desired by	All Customers			Average customer		Critical customer		Trained observer		Not perceptible

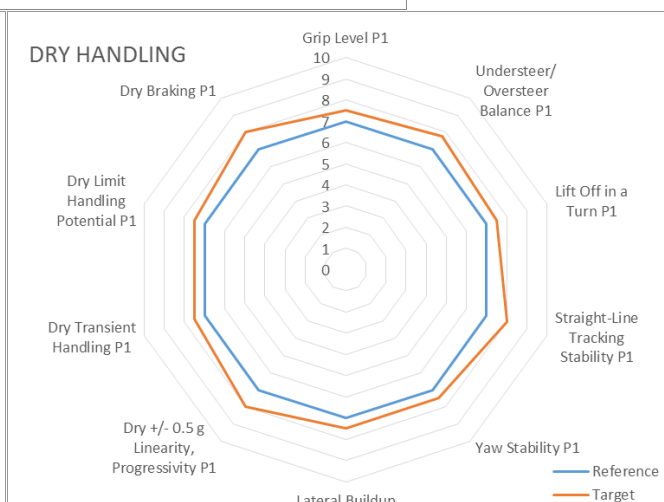
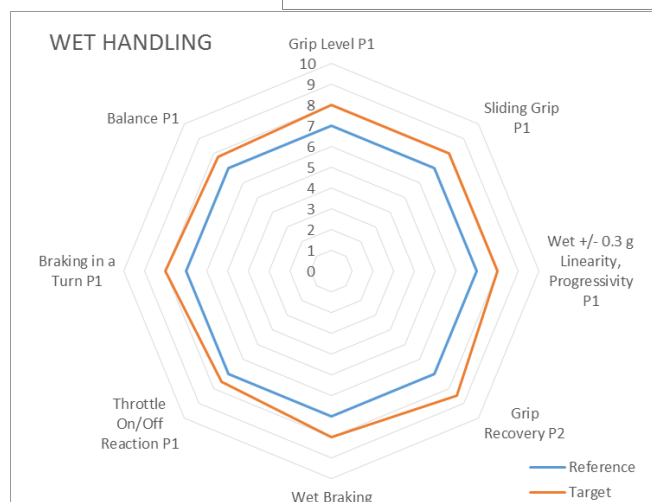
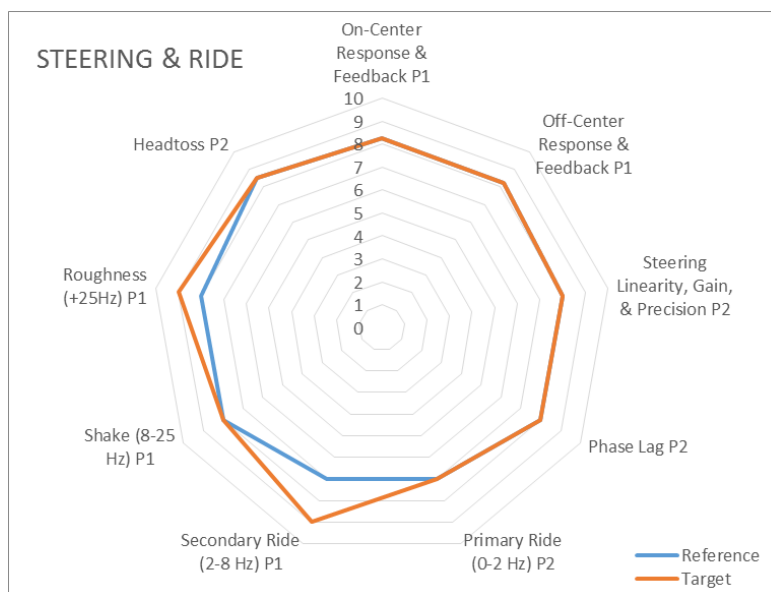


Figure 3: Sample Spider Chart. Chart shall include all Subjective Criteria from 4.6.1 as compared to a Reference Tire.

Current Reference for S1: Continental ProContact RX T0 (235/40R19)	Reference	TARGET	Ranking	Target Construction or Notes
Steering				
On-Center Response & Feedback				
Off-Center Response & Feedback				
Steering Linearity, Gain, & Precision				
Phase Lag				
Ride				
Primary Ride (0-2 Hz)				
Secondary Ride (2-8 Hz)				
Shake (8-25 Hz)				
Roughness (+25Hz)				
Headtoss				
Wet Handling				
Grip Level				
Sliding Grip				
Wet +/- 0.3 g Linearity, Progressivity				
Grip Recovery				
Wet Braking Traction				
Throttle On/Off Reaction				
Braking in a Turn				
Balance				
Dry Handling				
Grip Level				
Understeer/Oversteer Balance				
Lift Off in a Turn				
Straight-Line Tracking Stability				
Yaw Stability				
Lateral Buildup				
Dry +/- 0.5 g Linearity, Progressivity				
Dry Transient Handling				
Dry Limit Handling Potential				
Dry Braking				

4.7. Uniformity

Description	Per SAE J332, Test 4.7.1 thru 4.7.6
Parameters	Rim Width: Per Axle Load: GVW Per 2.3 Inflation Pressure: Per 2.3

Item	Description	Criteria	Markings
4.7.1	Conicity	± 70 N	1 stripe across the tread for the negative class 2 stripes across the tread for the positive class
4.7.2	Radial Run-out	± 1 mm	
4.7.3	Radial Force Peak to Peak (RFPP)	105 N	
4.7.4	Radial First Harmonic (RF1H)	75 N	Red paint dot at the high point
4.7.5	Lateral Force Peak to Peak (LFPP)	80 N	
4.7.6	Lateral Run Out	± 1 mm	

4.7.1 Inflation Pressure Loss Rate

Description	Test Method: ASTM F-1112
Parameters	Rim width and inflation pressure: Per Section 2.3
Criteria	Target: Pressure loss $\leq 1.5\%$ Max allowable: 2.0%

4.7.2 Flat Spotting – Short Term

Description	Measure change in tire uniformity due to short term chargings flat spotting		
Set Up	Step		Short Term
	Initial Uniformity Measurement	Measure RFPP and RFH1 per 4.7 Uniformity	n/a
	Warm Up and Measurement		
	Speed (kph)		100
	Inflation (kpa)		290
	Load		Performance Load, Rear
	Duration (min)		45
	Frequency (Sec.)		30
	Flatspot		
	Inflation (kpa)		310
	Load		Curb Load, Rear
	Duration (hours)		1
	Temp (°C)		50
	Cooling		
	Inflation (kpa)		n/a
	Load		n/a
	Duration (hours)		n/a
	Temp		n/a
	Reduction and Measurement		
	Speed (kph)		100
	Inflation (kpa)		290
	Load		Performance Load, Rear
	Duration (min)		45
	Frequency (Sec.)		30
Image	TBD % Increase in Uniformity as measured after reduction in Short Term Test		
Criteria	Reference values in 4.7 Uniformity, change from nominal and recovery time to be confirmed		

4.7.3 Flat Spotting – Long Term

Description	Measure change in tire uniformity due to Long Term parking flat spotting			
Set Up	Step		Nominal Temp Test	Elevated Temp Test
	Initial Uniformity Measurement	Measure RFPP and RFH1 per 4.7 Uniformity	Per cell temp	n/a
	Warm Up and Measurement			
	Speed (kph)		100	100
	Inflation (kpa)		290	290
	Load		Performance Load, Rear	
	Duration (min)		45	45
	Frequency (Sec.)		30	30
	Flatspot			
	Inflation (kpa)		290	290
	Load		Curb Load, Rear	
	Duration (hours)		24	6
	Temp (°C)		20	45
	Cooling			
	Inflation (kpa)		n/a	290
	Load		n/a	Curb Load, Rear
	Duration (hours)		n/a	2
	Temp		n/a	20
	Reduction and Measurement			
	Speed (kph)		100	100
	Inflation (kpa)		290	290
	Load		Performance Load, Rear	
	Duration (min)		45	45
	Frequency (Sec.)		30	30
Image				
Criteria	TBD % Increase in Uniformity as measured after reduction in Nominal Temp Test TBD % Increase in uniformity as measured after reduction in Elevated Temp Test			

5 Market Countries

Zone	Country
RENA	United States
RENA	Canada
RENA	Mexico
REAP	Australia
REAP	China
REAP	Hong Kong
REAP	Japan
REAP	New Zealand
REAP	Macau
REAP	South Korea
REAP	Taiwan
REEU	Austria
REEU	Belgium
REEU	Denmark
REEU	Finland
REEU	France
REEU	Germany
REEU	Ireland
REEU	Jordan
REEU	Luxembourg
REEU	The Netherlands
REEU	Norway
REEU	Portugal
REEU	Sweden
REEU	Switzerland
REEU	United Arab Emirates
REEU	United Kingdom

6 Appendix A: Referenced Procedures and Specifications

The following documents were referenced in the creation of this document. If there is a need for clarification or detail beyond the scope of this document, the following documents shall be used as a guideline for such issues.

- SAE J2047 - *Tire Performance Terminology*
- SAE J2452 - *Stepwise Coastdown Methodology for Measuring Tire Rolling Resistance*
- SAE J332 - *Testing Machine for Measuring the Uniformity of Passenger Car and Light Truck Tires*
- ISO 362 - *Pass-by noise testing standard*

7 Appendix B: Program Timing

Projected milestones, including submissions, vehicle builds, and validation requirements. Initial timing subject to change pending program schedule updates.

Mule development (Sub 0) development availability from Jan 2019

Joint Development (Sub 1) June 2019

Joint Development (Sub 2) Oct 2019

SOP June 2020